

Asian Call Option Talbot implementation

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0.1. Price according to Black-Scholes

Asian options' payout function is:

In the case of asian call:

$$\max\left(\frac{1}{T} \int_0^T S(t)dt - K, 0\right)$$

In the case of asian put:

$$\max\left(K - \frac{1}{T} \int_0^T S(t)dt, 0\right)$$

I will price the asian call option.

$$\frac{1}{T} \int_0^T S(t)dt - K$$

According to modern financial math's results, if we denote S as the price process of a financial instrument, and the markets are arbitrage free, there exist uniquely a measure (let's say) Q , under which S is a local martingale. We will call it martingale measure.

We define the Radon-Nikodym process, Λ , as

$$dQ/dP$$

, where P is the real world measure.

Under Real world measure, this financial instrument's price's SDE is:

$$dS = S\mu(t, \omega)dt + S\sigma(t, \omega)dW$$

One can show that, under Black-Scholes, (where expected value, variance, and risk free interest rate are constant) this Radon-Nikodym process can be written as

$$e^{\theta \bullet W - \frac{1}{2}[\theta \bullet W]} = e^{-\theta W(t) - \frac{1}{2}\theta^2 T}$$

where

$$\theta = \frac{\mu - r}{\sigma}$$

Which means, the Wiener-process under Q according to real world measure P is

$$E^Q(\chi_A W(t)) = \int_A W(t) dQ = \int_A W(t) \frac{dQ}{dP} dP = E^P(\chi_A W(t) \Lambda(t))$$

$$\begin{aligned} E^P(\chi_A W(t) e^{-\theta W(t) - \frac{1}{2}\theta^2 T}) &= \frac{1}{\sqrt{2\pi t}} \int_A e^{\frac{-x^2}{2t}} e^{-\theta x - \frac{1}{2}\theta^2 T} dx = \frac{1}{\sqrt{2\pi t}} \int_A e^{-\frac{1}{2}(\frac{x^2}{t} - 2\theta x + \theta^2 t)} dx \\ &= \frac{1}{\sqrt{2\pi t}} \int_A e^{-\frac{1}{2}(\frac{x}{\sqrt{t}} - \theta\sqrt{t})^2} dx = \frac{1}{\sqrt{2\pi t}} \int_A e^{-\frac{(x-\theta t)^2}{2t}} dx = N(\theta t, \sqrt{t}) \end{aligned}$$

Which means that the increments under P are based on this Normal distribution, where it is a Wiener process under Q

$$\widehat{W(t)} = W(t) - \theta t$$

so,

$$W(t) = \widehat{W(t)} + \theta t$$

is normal distribution with expected value of 0, and variance of t. We could show that this is wiener process under Q, based on independent increments. From this, under Q measure we get:

$$\begin{aligned} dS &= S\mu(t, \omega)dt + S\sigma(t, \omega)dW = S\mu(t, \omega)dt + S\sigma(t, \omega)(\widehat{w(t)} + \theta t) \\ &= S\mu(t, \omega)dt + S\sigma(t, \omega)(\widehat{w(t)}) - S\sigma(t, \omega)\theta dt = S\mu(t, \omega)dt + S\sigma(t, \omega)(\widehat{w(t)}) - S\sigma\frac{\mu - r}{\sigma}dt = \\ &= dS = Srdt + S\sigma d\widehat{W} \end{aligned}$$

Now we know S because

Using Ito's lemma:

$$\begin{aligned} S &= e^{\ln(S)} \Rightarrow \ln S = \frac{df}{dt} \bullet t + \frac{df}{dS} \bullet S + \frac{1}{2} \frac{ddf}{dS dS} d[S] = \\ &= \frac{1}{S} dS + \frac{1}{2} \frac{1}{S^2} d[S] \quad dS = Srdt + S\sigma d\widehat{W} \quad \frac{1}{S} (Srdt + S\sigma d\widehat{W}) + \frac{1}{2} \frac{1}{S^2} S^2 \sigma^2 dt \end{aligned}$$

from here:

$$S = e^{\ln(S)} = e^{(r - \frac{1}{2}\sigma^2 dt + \sigma d\widehat{W})}$$

So inserting S into the Riemann integral:

$$\int_0^T S(t) dt = S(0) \int_0^T \exp \left(\left(r - \frac{\sigma^2}{2} \right) t + \sigma \widehat{w}(t) \right) dt =$$

Now it's a bit tricky, because we have to define t as $t := 4u / \sigma^2$

So now it looks like:

$$= S(0) \frac{4}{\sigma^2} \int_0^{\sigma^2 T / 4} \exp \left(\left(r - \frac{\sigma^2}{2} \right) \frac{4}{\sigma^2} u + \sigma \widehat{w} \left(\frac{2}{\sigma} \frac{2}{\sigma} u \right) \right) du \quad \widehat{w}(ct) \stackrel{=}{=} \sqrt{c} \widehat{w}(t)$$

Let's introduce

$$\nu := \frac{2r}{\sigma^2} - 1$$

Now

$$= S(0) \frac{4}{\sigma^2} \int_0^{\sigma^2 T / 4} \exp(2(\nu + \widehat{w}(u))) du$$

If we define

$$\Psi(T, K, \nu) := \mathbb{E}^Q \left(\left(\int_0^T \exp(2(\nu \cdot u + \widehat{w}(u))) d\mu - K \right)^+ \right)$$

then

$$\pi(H_T) = \exp(-rT) \mathbb{E}^Q \left(\frac{S(0)}{T} \frac{4}{\sigma^2} \int_0^{\sigma^2 T/4} \exp(2(\nu \cdot u + \widehat{w}(u))) du - K \right)^+$$

after bringing out $\frac{S(0)}{T} \frac{4}{\sigma^2}$ from integral (which in this case is the expected value)

$$= \exp(-rT) \frac{S(0)}{T} \frac{4}{\sigma^2} \mathbb{E}^Q \left(\int_0^{\sigma^2 T/4} \exp(2(\nu \cdot u + \widehat{w}(u))) du - \frac{KT\sigma^2}{4S(0)} \right)^+$$

Inserting Ψ using it's definition:

$$= \exp(-rT) \frac{S(0)}{T} \frac{4}{\sigma^2} \Psi \left(\frac{\sigma^2 T}{4}, \frac{KT\sigma^2}{4S(0)}, \nu \right).$$

So now we "only" have to calculate $\Psi(T, K, \nu)$

We will do this by Using German- Yor's theorem

German-Yor theorem

For any $\lambda > \max(0, 2(\nu + 1))$

$$L(\lambda, T, K, \nu) = \int_0^\infty \exp(-\lambda T) \Psi(T, K, \nu) dT = \frac{(2K)^{1-\beta}}{2\lambda(\alpha+1)\Gamma(\beta)} \int_0^1 u^{\beta-2} (1-u)^{\alpha+1} \exp\left(-\frac{u}{2K}\right) du,$$

where

$$\gamma \doteq \sqrt{2\lambda + \nu^2}, \quad \alpha \doteq \frac{\gamma + \nu}{2}, \quad \beta \doteq \frac{\gamma - \nu}{2}.$$

The proof is very long, so I will not prove it. I got it from Peter Medvegyev's "Pénzügyi Matematika" book.

But from Wikipedia we know that

$$M(a, b, x) = \frac{\Gamma(b)}{\Gamma(a)\Gamma(b-a)} \int_0^1 \exp(xu) u^{a-1} (1-u)^{b-a-1} du.$$

Where M is Kummer's function of a generalized hypergeometric series:

$$M(a, b, z) = \sum_{n=0}^{\infty} \frac{a^{(n)} z^n}{b^{(n)} n!} = {}_1F_1(a; b; z),$$

where

$$\begin{aligned} a^{(0)} &= 1, \\ a^{(n)} &= a(a+1)(a+2) \cdots (a+n-1), \end{aligned}$$

So, computationally, for each n can be easily calculated by the next easy formula:

$$\frac{a^{(n)} z^n}{b^{(n)} n!} = \frac{a^{(n-1)} z^{n-1}}{b^{(n-1)} (n-1)!} * \frac{(a+n)z}{(b+n)n}$$

So now we know that the Laplace-transform can be expressed by

$$\begin{aligned} L(\lambda, T, K, \nu) &= \frac{(2K)^{1-\beta}}{2\lambda\Gamma(\beta)(\alpha+1)} \int_0^1 \exp\left(-\frac{u}{2K}\right) u^{\beta-2}(1-u)^{\alpha+1} du = \\ &= \frac{(2K)^{1-\beta}}{2\lambda\Gamma(\beta)(\alpha+1)} \frac{\Gamma(\beta-1)\Gamma(\alpha+2)}{\Gamma(\alpha+\beta+1)} M\left(\beta-1, \alpha+\beta+1, -\frac{1}{2K}\right) = \\ &= \frac{(2K)^{1-\beta}}{2\lambda(\beta-1)} \frac{\Gamma(\alpha+1)}{\Gamma(\alpha+\beta+1)} M\left(\beta-1, \alpha+\beta+1, -\frac{1}{2K}\right). \end{aligned}$$

0.2. Numerical inverse Laplace-transformation

By looking at the Laplace transform's definition it is easy to see it is hard to calculate the inverse numerically.

$$L(\lambda, T, K, \nu) = \int_0^\infty \exp(-\lambda T) \Psi(T, K, \nu) dT$$

We have to use Mellin's inverse formula the Bromwich integral:

$$\int_{\sigma-i\infty}^{\sigma+i\infty} e^{st} F(s) ds,$$

We have to add to this a semicircle, to close the contour, because we want to implement a new contour, which has an integral on a computationally easier set, by Cauchy's residue theorem.

$$\oint_{\Gamma_R} e^{st} F(s) ds = \int_{\sigma-iR}^{\sigma+iR} e^{st} F(s) ds + \int_{C_R} e^{st} F(s) ds.$$

Where $R > 0$.

Now let's define a closed semicircle:

$$\Gamma_R = (\sigma - iR \rightarrow \sigma + iR) \cup C_R,$$

where

$$C_R : s(\theta) = \sigma + Re^{i\theta}, \quad \frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}.$$

If there exist constants $M > 0, k > 0$ such that on C_R (where $s = Re^{i\theta}$)

$$|f(s)| < \frac{M}{R^k},$$

then the limit of the integral of $e^{st} f(s)$ along C_R as $R \rightarrow \infty$ is zero, that is,

$$\lim_{R \rightarrow \infty} \frac{1}{2\pi i} \int_{C_R} e^{st} f(s) ds = 0.$$

It can be shown that the condition is always satisfied, since in the Geman-Yor formula

$$\Gamma(\beta) = \Gamma\left(\frac{\sqrt{2\lambda + \nu^2} - \nu}{2}\right),$$

which means

$$|\Gamma(Re^{i\theta})| = \left| \int_0^\infty t^{Re^{i\theta}-1} e^{-t} dt \right| > R^k$$

This is what needs to be satisfied. And it is, because the Gamma function satisfies this, it grows very fast. Since $3\pi/2 > \theta > \pi/2$, the integrand is negative, but due to the absolute value THE INTEGRAL IS POSITIVE, and in the Geman-Yor formula the gamma is in the denominator, meaning the theorem holds for it !!!! .

Since

$$\oint_{\Gamma_R} e^{st} F(s) ds = \int_{\sigma-iR}^{\sigma+iR} e^{st} F(s) ds + \int_{C_R} e^{st} F(s) ds.$$

It is closed, so we can switch to an integral along another closed contour. (Here I would like to note that due to the conditions and results of the Geman-Yor theorem, the analyticity of the Laplace transform on the right half-plane is a given because by German-Yor theorem's definition there can be no singularities on the right half plane). Now we can switch to the Talbot contour. This is the so-called Talbot contour:

$$\Gamma : z(\theta) = \sigma + \mu(\theta \cot \theta + \nu i \theta), \quad -\pi \leq \theta \leq \pi,$$

Then

$$\int_{\Gamma} e^{zt} L(z) dz = \int_{-\pi}^{\pi} e^{z(\theta)t} L(z(\theta)) z'(\theta) d\theta.$$

That is, putting it all together: The Laplace transform:

$$L(\lambda, T, K, \nu) = \int_0^\infty \exp(-\lambda T) \Psi(T, K, \nu) dT$$

So, by applying Mellin inverse Laplace transform, we get the Bromwich integral:

$$\begin{aligned} \Psi(T, K, \nu) &= \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} e^{\lambda T} L(\lambda) d\lambda = \\ R \rightarrow \infty : \frac{1}{2\pi i} \left(\int_{\sigma-iR}^{\sigma+iR} e^{\lambda t} L(\lambda) d\lambda + \int_{C_R} e^{\lambda t} L(\lambda) d\lambda \right) &= \frac{1}{2\pi i} \oint_{\Gamma_R} e^{\lambda t} L(s) d\lambda = \\ \frac{1}{2\pi i} \int_{\Gamma} e^{zt} L(z) dz &= \frac{1}{2\pi i} \int_{-\pi}^{\pi} e^{z(\theta)t} L(z(\theta)) z'(\theta) d\theta. \end{aligned}$$

We know now the Laplace-transform, function $z(\theta)$, we only need $z'(\theta)$.

$$\begin{aligned} z'(\theta) &= f'g + g'f \stackrel{g=\frac{u}{v}}{=} f'g + \frac{u'v - v'u}{v} f = \frac{1}{\tan(\theta)} + \theta \frac{-\sin(\theta)\sin(\theta) - (-\cos(\theta)\cos(\theta))}{\sin^2(\theta)} \\ &= \frac{1}{\tan(\theta)} - \theta \frac{\sin^2(\theta) + \cos^2(\theta)}{\sin^2(\theta)} = \mu \left(\frac{1}{\tan(\theta)} - \frac{\theta}{\sin(\theta) * \sin(\theta)} + \nu * i \right) \end{aligned}$$